

# Yiğit Sarp Avcı

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## Education

**Boğaziçi University - B.Sc. in Computer Engineering**

09/2023 – Present

**Kabataş Boys' High School**

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00; Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.

## Experience

**Executive Board Member at Radyo Boğaziçi**

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

**Work and Travel Program Participant**

06/2025 – 09/2025

- Gained international work experience in the USA, improving English communication, teamwork, and adaptability in a multicultural environment.

**Regional Organizational Team Member at ESTIEM**

09/2023 – Present

- Contributed to the planning, coordination of events, and organizational processes within ESTIEM, an international network for Industrial Engineering and Management students.

## Technical Projects

**AlphaTrigger AI Terminal** — Python, FastAPI, Next.js, WebSockets

[GitHub Link](#)

- Autonomous high-frequency trading simulation engine featuring deterministic risk-management strategies, dynamic volatility-adjusted position sizing, and a real-time WebSocket-driven dashboard.

**Performo Enterprise Admin** — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

[GitHub Link](#)

- Developed an administrative dashboard with real-time operational metrics and server-side synchronized state management for system monitoring; implemented a design system with sub-second data refresh cycles via force-dynamic rendering and optimized SQL indexing for database stability.

**AI-Powered FinTech Intelligence Platform** — Next.js, FastAPI, PostgreSQL, Docker

[GitHub Link](#)

- Built a full-stack, deployment-ready financial dashboard combining personal finance analytics, a portfolio risk evaluator, and OpenAI-generated insights with secure NLP fallbacks; engineered a live limit-order trading simulator featuring a synchronous Price-Time Priority matching engine with automated CI/CD pipelines and high test coverage.

**SchedSim CPU Scheduler** — x86-64 Assembly, GNU Toolchain, Linux Syscalls

[GitHub Link](#)

- High-performance CPU scheduling simulation engine (FCFS, SJF, SRTF, PF, RR) built with zero standard library dependencies, utilizing direct Linux syscalls for I/O and process management.

**Spirelayer Command Interpreter** — C, Custom Backtracking Lexer

[GitHub Link](#)

- CLI-based interpreter replicating fantasy RPG REPL mechanics; implemented an autonomous backtracking lexical parser to handle non-deterministic grammar and command resolution.

**Aura Functional Interpreter** — Python, Functional Paradigms, Dual Scoping Engine

[GitHub Link](#)

- Interpreter for a custom functional language supporting first-class closures, higher-order function patterns, and toggleable lexical/dynamic scoping engines.

**MatrixNet** — Java, Tarjan's SCC Algorithm, Graph Theory

[GitHub Link](#)

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection ( $O(V + E)$ ); optimized with Open Addressing to eliminate internal JVM string allocation overheads.

**GigMatch-Pro** — Java, Memory-Optimized Heaps

[GitHub Link](#)

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for priority allocation and candidate-to-job matching algorithms.

**Nightpass Survival Game** — Java, Rule Engine Architecture

[GitHub Link](#)

- Algorithmic implementation of a rule engine governing strategic deck, hand, and resource economics in a terminal-based survival card game.

**Qt 2048 WebAssembly** — C++, Qt6, WebAssembly (WASM)

[GitHub Link](#)

- Engineered a high-performance clone of the 2048 game using C++ and the Qt6 framework; cross-compiled for the web via WebAssembly (WASM) for high-fidelity browser execution with optimized grid merging logic and infinite state-history for undos.

## Technical Skills

- **Programming:** Python, Java, C/C++, Qt6, WebAssembly (WASM), TypeScript, JavaScript, SQL, x86-64 Assembly, Bash
- **Low-Level & Systems:** Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management, Qt Framework
- **Enterprise Full-Stack:** Next.js 15+, React, Prisma (ORM), SQLAlchemy (ORM), FastAPI, Tailwind CSS, Framer Motion, PostgreSQL, SQLite
- **Data Structures & Algos:** Tarjan's SCC, Articulation Points, Max-Heaps, Complexity Analysis (Big O), Graph Theory, Recursive Optimization
- **Paradigms & Models:** Functional (Scheme/Racket), Logic (Prolog), OOP (C++/Qt), Interpreter Design, State Machines
- **AI & Modern Engineering:** OpenAI API Integration, Advanced Agentic Coding, LLM Prompting, AI-Assisted Architecture, Autonomous Feature Deployment, Data Analysis Basics
- **Tools & Ecosystem:** Linux/Unix, Git/GitHub, Docker, Alembic, ccache, Pytest (Testing), CI/CD, Makefile, LaTeX, Adobe Photoshop, Microsoft Excel, PowerPoint, Canva

## Volunteer Experience

- **Derece Kampüsü:** Mentored high school students for university entrance exams, providing academic guidance. (09/2023 – 09/2025)
- **TOÇEV:** Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds. (09/2019 – 03/2020)
- **Boğaziçi University Industrial Engineering Club:** Actively contributed to the Volunteering Division. (09/2023 – 06/2024)
- **Kabataş Boys' High School CIP Club:** Organized and executed various community-oriented projects. (09/2018 – 06/2023)

## Languages

- Turkish (native)
- English (fluent)
- German (elementary)

## Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually